RV UNIVERSITY
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ANVAYA 2k26
IDEA IMPACT · BUSINESS PITCHING

TOUCH-ACTIVATED PERSONAL SAFETY

SparshaGuard

Sparśa · Kannada for touch

The touch itself raises the alarm.

A 4G smartwatch that calls her family and 112 when a stranger grabs her. No phone. No button.

- Own 4G SIM
- 15-second PIN
- Duress PIN



Bhajanth V
B.Tech CSE (GC) · RV University

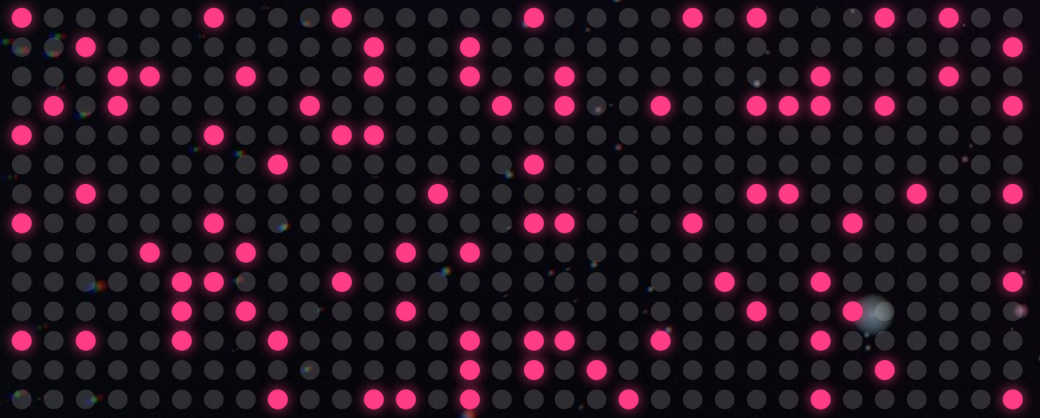
Sukruth A Chinnur
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01 · THE PROBLEM

Every SOS tool needs her to act first.

4,48,211

crimes against women recorded in India in 2023



● = 1,000 cases ● assault to outrage her modesty: 83,891, about 1 in 5

51
every hour, around the clock

40%
of urban women feel unsafe or not so safe

TO CALL FOR HELP TODAY, SHE MUST

1 Find the phone → 2 Unlock it → 3 Open the app → 4 Press SOS

During an attack, her hands are not free, and the phone is out of reach.

NCRB, Crime in India 2023 (released Sep 2025): 4,48,211 cases; Sec. 354 IPC 83,891. NARI 2025 (National Commission for Women): 12,770 women, 31 cities.

02 · THE SOLUTION

The watch notices the attack by itself.

- Her own hand: ignored
- Watch ripped off: instant alert



Guard Mode on

She switches it on when she feels unsafe. Otherwise, it is a normal smartwatch.



A stranger's touch → 15-second PIN

A grab, a pull or a struggle. It vibrates, starts recording and asks for her PIN.



No PIN → family + 112 alerted

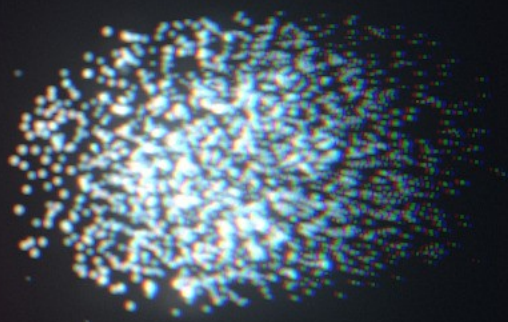
Her live GPS location and audio go out over its own 4G SIM.



Forced to cancel? Duress PIN

The screen says "Cancelled" while the alert is sent silently.





03 · HOW IT KNOWS

Two signals tell a stranger from her own hand.

SIGNAL 1 · TOUCH SWEEP

Is a hand on the strap?

A tiny signal sweeps through the strap. Any hand on it moves the reading by **~16%**, against **0.1%** noise.

SIGNAL 2 · STATIC CHARGE

Is it another body?

Every body carries its own static charge. When another person touches her, it evens out and her voltage **jumps**.

	TOUCH SWEEP	STATIC CHARGE	THE WATCH
Her own hand	changes	flat: same body	stays quiet
A stranger	changes	jumps	asks for PIN
A struggle	—	arm jerks (motion)	asks for PIN



Built on **60** published papers

Cross-checked by **2** independent solvers: ngspice, MIT FastCap

04 · THE PRODUCT

A flagship 4G smartwatch, with a safety core inside.

- 📶 Own nano-SIM: 4G VoLTE calls and SMS
- 📍 Built-in GPS, live location
- ⌚ 1.43″ AMOLED, always-on, crown
- ❤ Heart rate · SpO_2 · ECG · skin temperature
- ⇄ Wi-Fi 6 · Bluetooth 5.3 · NFC
- ⚡ 500 mAh · wireless charging

Two processors. One runs the apps; a separate **safety core** runs Guard Mode alone, so a frozen app can never block an alert. Our own SparshaOS, on open-source Zephyr.



Sapphire glass

Ceramic bezel

1.43″ AMOLED · touch screen

Main board · safety core + app processor

Titanium case · crown, buttons

Cavli C11QM · 4G VoLTE + GPS

500 mAh battery

Sensor board · heart rate · ECG · static

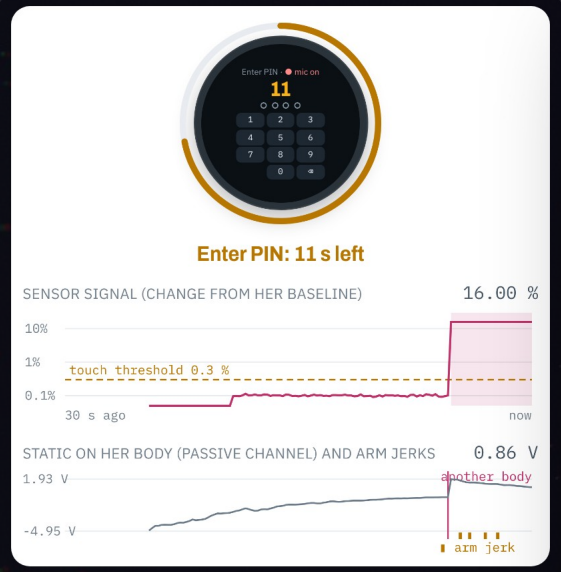
Steel case back · skin electrode

● Design specification · hardware prototype next

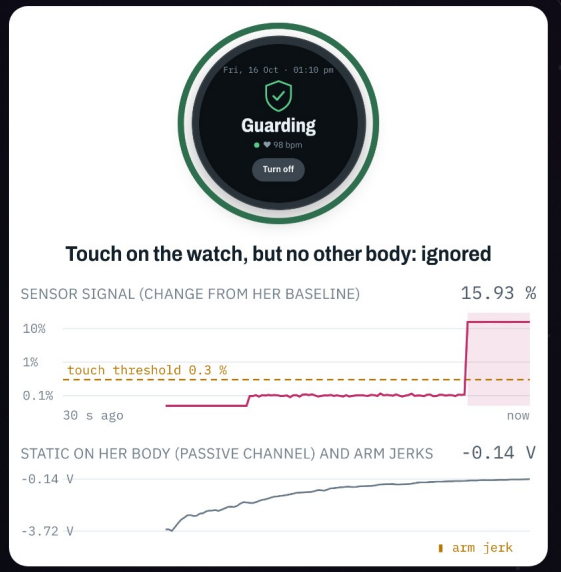
05 · PROOF

Tested on a digital twin before any hardware.

The watch's real software, running on a simulated body. Live on our website.



● A stranger grabs her wrist: both signals jump → **PIN**



● Her own hand: static stays flat → **ignored**

70%
of grabs caught across 7 places on her body
(strap sweep alone: 13%)

99%
of struggles caught by the motion sensor

0–1
unwanted PIN prompts in 9 simulated hours

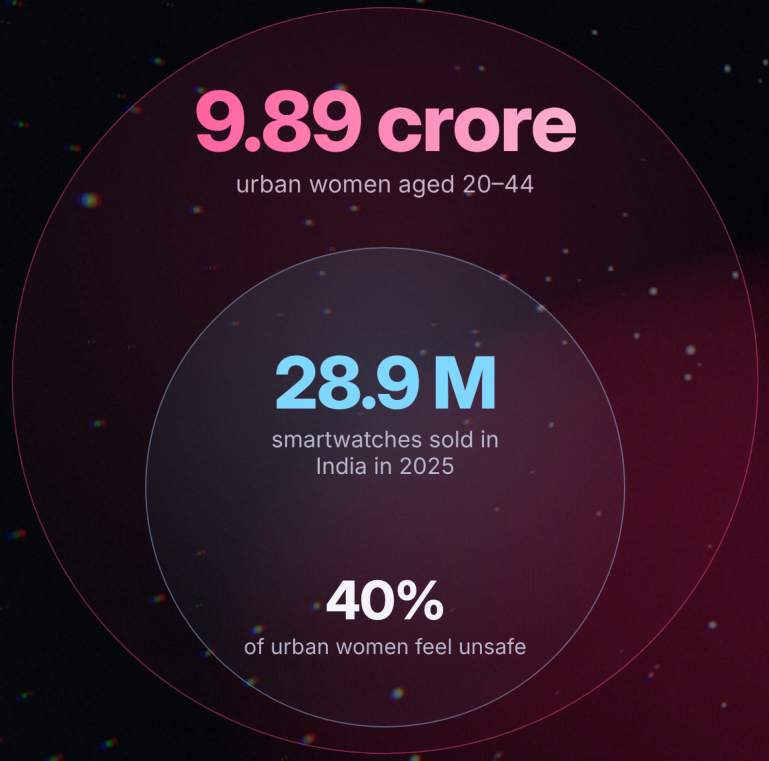
100%
of strap removals caught

● bhajanth13.github.io/SparshaGuard

Simulation results, not yet measured on hardware. The Gate 1 battery prototype will calibrate them against real body voltages.

06 · MARKET & COMPETITION

A huge market, and no one detects the touch.



PRODUCT	PRICE	WORKS WITHOUT A PHONE	RAISES THE ALARM BY ITSELF
Budget SOS watches Noise, boAt	₹1,999–2,699	✕ needs the phone	✕ press a button
Children's 4G watches Pebble Junior	₹5,999	✓ own SIM	✕ press SOS
Premium LTE watches Apple, Samsung	₹41,999–51,499	✓ own LTE	◖ only a hard fall or crash
SparshaGuard Band · Watch	**₹11,999 · ₹24,999**	✓ own 4G SIM	**✓ another person's touch, a pull or a struggle**

The closest idea is an MIT research sticker, never sold. **No product in India raises an alert on another person's touch.**

MoSPI, Women and Men in India (population projection); IDC India wearables, 2025; NARI 2025. Prices as listed by brands and retailers.

07 · BUSINESS MODEL

Two products, four revenue streams, one free promise.

SPARSHAGUARD BAND

₹11,999

The safety system alone: 4G, GPS, PIN screen, sensing.

Making cost ≈ ₹6,500*

SPARSHAGUARD WATCH

₹24,999

The full flagship smartwatch, with Guard Mode built in.

Making cost ≈ ₹17,000 → ₹11,500 at scale*

GUARD+ FAMILY APP

₹99 /month

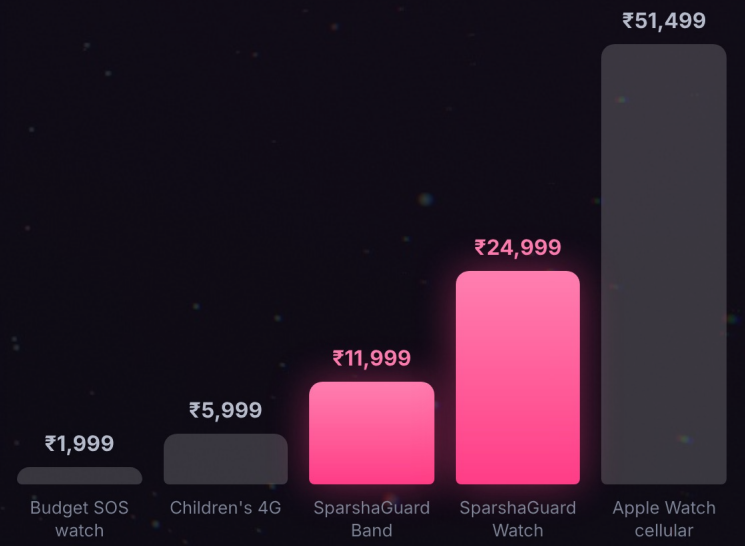
Live map during an alert, encrypted recordings, alert history.

CAMPUS & WORKPLACE

₹4,500 /user/year

Band, Guard+ and replacements, a safety dashboard, 3-year term.

PRICE LADDER · BARS TO SCALE



- Calling for help: free for life
- We never sell location or audio data

* Our estimates from typical distributor prices at 1,000 units (₹11,500 at 10,000 units), to be confirmed with supplier quotes. Prices include GST.

08 · GO-TO-MARKET & ROADMAP

Start where safety is bought in bulk.

PHASE 1 · YEAR 1

Campus pilot

200 users in colleges and hostels in Karnataka, starting at RV University.

Paid by the institution or parents

PHASE 2 · YEAR 2

Employers

IT, BPO, healthcare and retail firms with women on night shifts.

Paid by the employer

PHASE 3 · YEAR 3

Online and retail

Direct to women and their families, on our site and marketplaces.

Paid by the customer

HARDWARE ROADMAP

DONE
Simulation study + digital twin
60 papers · 2 solvers

1
GATE 1
Battery bench prototype
Measures real body voltages

2
GATE 2
Wearable SparshaGuard Band
Custom SMT board, own SIM

3
GATE 3
Full SparshaGuard Watch
Two processors, AMOLED

✦
LAUNCH
Indian certification
WPC · TEC/MTCTE · BIS

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Thank you.

SparshaGuard: the touch itself raises the alarm.

We welcome your questions.



PRESENTED BY

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Try the digital twin
bhajanth13.github.io/SparshaGuard